1 / THE PRODUCTS

OSMO connects tasks into a workflow

Use OSMO when several container tasks need to work together.

It handles their resources, dependencies and data.

Simulate
Generate robot data

→

Train
Build a model

→

Evaluate
Check the model

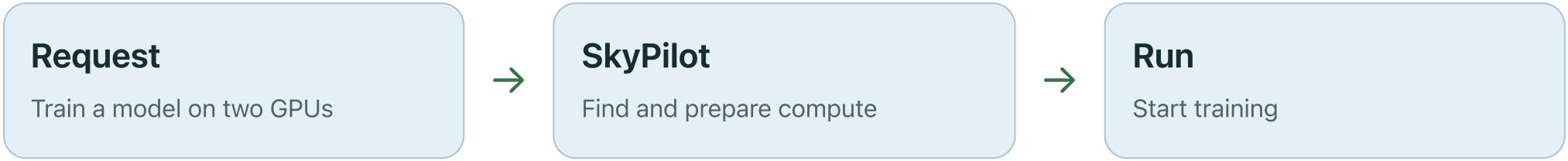
> The output of one task becomes the input of the next.

1 / THE PRODUCTS

SkyPilot finds compute and runs your job

Use SkyPilot to run a task across clouds or Kubernetes.

It chooses suitable compute from the infrastructure you allow and prepares the environment.



> These resources form a SkyPilot compute cluster. It can run another job later.

2 / THE METRICS GAP

We want to see usage for that work

This example covers workloads running as Pods in Kubernetes.

OSMO

Workflow → tasks

How much GPU did this workflow and its tasks use?

SKYPILOT

Compute cluster → roles

How much GPU did this compute cluster and its head / worker roles use?

> To answer, connect each measured Pod to its platform grouping.

1. Recorded metadata association

2 / THE METRICS GAP

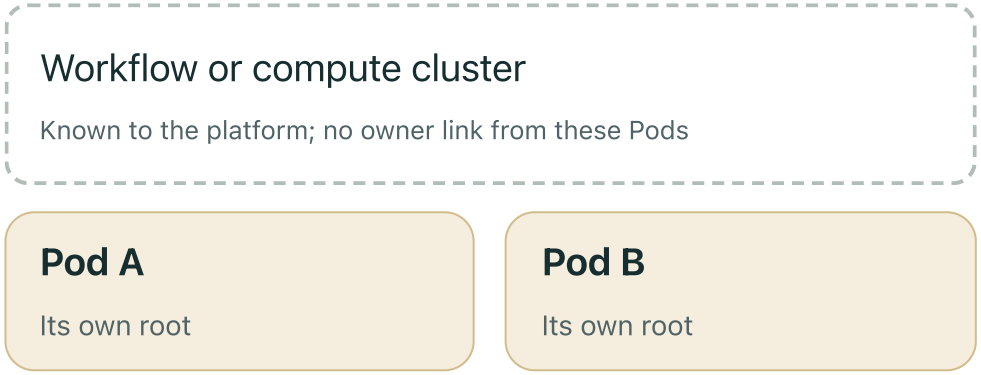
The platform grouping is outside Kubernetes ownership

A KUBERNETES WORKLOAD



Karta follows the owner links to the JobSet.

THE CAPTURED OSMO / SKYPILOT PODS



With the generic Pod definition, Karta reports each Pod separately.

> Karta sees the Pods. It cannot infer their common platform grouping.

1. OSMO captured objects 2. SkyPilot captured objects 3. Karta ownership contract

2 / THE METRICS GAP

Illustrative values

The Pods already carry useful labels

OSMO POD

`osmo.workflow_uuid`

W

`osmo.task_name`

train

Group by workflow.
Then show each task.

SKYPILOT POD

`skypilot-cluster-name`

C

`ray-node-type`

head

Group by compute cluster.
Then show each role.

Short example IDs. Scope each key to its platform installation and Kubernetes cluster / namespace.

1. OSMO labels
2. SkyPilot labels

3 / WAYS TO SOLVE IT

Available approach

Option 1: join the metadata in Prometheus

Enable the needed Pod labels in kube-state-metrics.

Join them to Karta's Pod metrics by namespace and Pod UID.

Karta Pod mapping + Pod labels → **Prometheus** Join the same Pod → **Grouped view** Workflow or compute cluster

Include Kubernetes cluster identity when querying several clusters.

> This metadata join worked in the lab. Each Pod keeps its real UID.

1. Tested metadata join rules
2. Recorded results

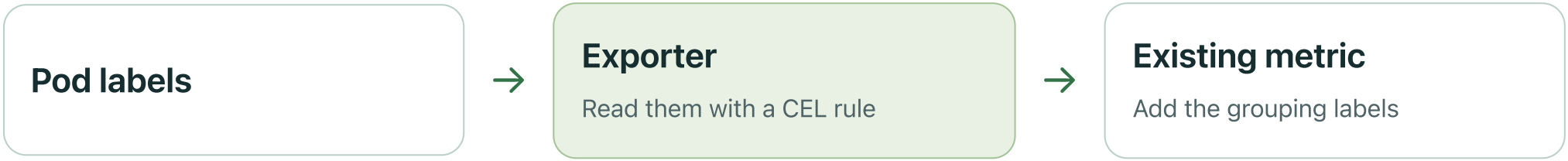
3 / WAYS TO SOLVE IT

Proposed

Option 2: let the exporter read the labels

Issue #518 proposes configurable CEL expressions.

CEL lets a rule read a field from a Pod or workload.



> The exporter would attach the grouping information directly.

Proposed work. It does not replace the join to GPU measurements.

1. Generic metric labels proposal #518

3 / WAYS TO SOLVE IT

Optional extension

We could also use Karta's component names

A further extension could map platform fields into Karta's existing attribution fields.

Example	component	component_instance	replica_group_id
OSMO training task	task	train	empty
SkyPilot head Pod	head	empty	empty
SkyPilot worker Pod	worker	empty	empty

Keep native Karta mappings first. Apply an explicit fallback only to generic Pod roots.

Leave the replica group empty unless the application defines a repeated group.

> Issue #518 currently forbids these overrides. This would extend it.

The workflow or cluster key is still needed to group the Pods.

1. Current #518 boundary 2. Existing attribution fields

3 / WAYS TO SOLVE IT

Labels do not answer every question

WORKFLOW STATUS

Ask the platform

Which tasks are waiting, running or complete? That needs platform state.

USAGE PER SKYPILOT JOB

Shared compute needs more detail

Two jobs can share one Pod. One Pod measurement cannot tell how much each job used.

> Karta supplies identity. DCGM supplies GPU measurements.

The lab verified metadata mapping, not GPU usage or successful SkyPilot job execution.

1. SkyPilot shared jobs
2. Recorded lab boundaries

3 / WAYS TO SOLVE IT

Start with the platform grouping

Now	Use the metadata join to group Pod identities today.
Next	Implement #518 to put grouping labels directly in exporter metrics.
If needed	Add component overrides for common component panels.

OSMO: workflow → task

SkyPilot: compute cluster → role

Each Pod keeps its real Kubernetes identity.

1. Metadata association approach 2. Exporter proposal